Diffusion Models for Combinatorial Optimization

Undergraduate Thesis

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Bachelor of Science in Industrial Engineering

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Abstract

Keywords: combinatorial optimization, traveling salesman problem, capacitated vehicle routing problem, diffusion models, reinforcement learning, neural combinatorial optimization.

Acknowledgments

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1.3 Contributions

- Reproduce and extend *DifusCO*-style diffusion solvers for TSP and CVRP.
- Implement *CADO*-style hybrid fine-tuning and REINFORCE on top of supervised checkpoints.
- Empirical comparison: optimality gap, wall-clock cost, and ablations (reward mode, entropy, denoising steps).

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